

Feature Processing and Selection

AI in Music, Winter Term 2025/2026

06.11.2025

Exercise 1 [Onset Detection, 3 points]

Load the classical music piece from last week's exercises in python.

Use the `librosa.onset.onset_detect` function to detect the onsets in the piece. Visualise the onsets together with the waveform.

Exercise 2 [Aggregation of Feature Values, 3 points]

Slice the file you loaded in Exercise 1 into non-overlapping windows of 5 seconds each. Then, load the AMUSE features you extracted last week and aggregate the features by calculating their mean inside each 5 second window.

Count how many onsets there are in each window and create a plot of feature values and the number of onsets in a window. (Y-axis: mean feature value, X-axis: Number of onsets in a window)

Do you notice a difference between windows with many onsets and windows with fewer onsets?

Exercise 3 [Selection of Time Frames, 4 points]

Within each 5 second window select only the frames that have onsets and, separately, the frames that are exactly in the middle between two onsets. For both of those cases, create the same plots from exercise 2.

Which features are the most affected by this selection of frames?